

✓ NLP chunk

change runtime to gpu | colab notebook -> file -> unmount drive

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
!nvidia-smi
```

/bin/bash: line 1: nvidia-smi: command not found

```
!pip install transformers
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.11/dist-packages (4.53.3)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from transformers) (3.18.0)
Requirement already satisfied: huggingface-hub<1.0,>=0.30.0 in /usr/local/lib/python3.11/dist-packages (from transformers) (0.33.4)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-packages (from transformers) (25.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packages (from transformers) (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.11/dist-packages (from transformers) (2024.11.6)
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from transformers) (2.32.3)
Requirement already satisfied: tokenizers<0.22,>=0.21 in /usr/local/lib/python3.11/dist-packages (from transformers) (0.21.2)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.11/dist-packages (from transformers) (0.5.3)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.11/dist-packages (from transformers) (4.67.1)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<1.0,>=0.30.0->transformers) (2025.7.1)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<1.0,>=0.30.0->transformers) (4.12.2)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.2 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub<1.0,>=0.30.0->transformers) (1.1.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests->transformers) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests->transformers) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests->transformers) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests->transformers) (2025.7.1)
```

```
from transformers import AutoTokenizer
```

```
tokenizer = AutoTokenizer.from_pretrained("gpt2")
```

```
text = "Hello, how are you?"
```

```
tokens = tokenizer(text,return_tensors = 'pt')
```

```
print(tokens)
```

```
{'input_ids': tensor([[15496, 11, 703, 389, 345, 30]]), 'attention_mask': tensor([[1, 1, 1, 1, 1, 1]])}
```

```
from transformers import AutoModelForCausalLM
```

```
model = AutoModelForCausalLM.from_pretrained("gpt2")
```

```
input_ids = tokenizer.encode("indian cricket", return_tensors = 'pt')
```

```
output = model.generate(input_ids, max_length = 50)
```

```
generated_text = tokenizer.decode(output[0],skip_special_tokens=True)
```

```
print(generated_text)
```

```
model.safetensors: 100% 548M/548M [00:06<00:00, 58.2MB/s]
```

```
generation_config.json: 100% 124/124 [00:00<00:00, 11.8kB/s]
```

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, you may observe unexpected behavior. Please pass your input Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

The team's captain, Ravi Shankar, has been in the country for over a decade.

The team's captain, Ravi Shankar,

```
from transformers import pipeline
```

```
generator = pipeline("text-generation",model = 'gpt2')
```

```
response = generator('What is Hugging Face',max_length = 40, num_return_sequences =1)
```

```
print(response[0]['generated_text'])
```

Device set to use cuda:0

What can I help you build?

+

Truncation was not explicitly activated but `max\_length` is provided a specific value, please use `truncation=True` to explicitly truncate the output. Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

Both `max\_new\_tokens` (=256) and `max\_length` (=40) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the

What is Hugging Face and What is Hugging Heart?

The goal of Hugging Face is to provide the best possible opportunity for your character to express their own personalities and emotions.

What are the main differences between Hugging Face and Heart?

Hugging Face is designed to be a family friendly game with a simple, fun game play experience. Hugging Heart is designed to be a fan game.

What do I need to know to play.

We use the Unity Asset Store to store all of the resources you need to play. Please check the Unity Asset Store for more information.

Do I have to be a parent or guardian?

While we don't have a "parent" requirement, if we have a child,

✓ NER(Named Entity Recognition)

```
#from transformers import pipeline
ner = pipeline('ner',model = 'dslim/bert-base-NER')
entities = ner('Hugging Face is based on NYC and partner with Google with $200 with dollar price')
print(entities)
```

```
config.json: 100% 829/829 [00:00<00:00, 35.6kB/s]

model.safetensors: 100% 433M/433M [00:04<00:00, 109MB/s]
Some weights of the model checkpoint at dslim/bert-base-NER were not used when initializing BertForTokenClassification: ['bert.pooler.dense.weight', 'bert.pooler.dense.bias']
- This IS expected if you are initializing BertForTokenClassification from the checkpoint of a model trained on another task or with a different architecture
- This IS NOT expected if you are initializing BertForTokenClassification from the checkpoint of a model that you expect to be exact copy of
tokenizer_config.json: 100% 59.0/59.0 [00:00<00:00, 3.98kB/s]

vocab.txt: 213k/? [00:00<00:00, 5.86MB/s]

added_tokens.json: 100% 2.00/2.00 [00:00<00:00, 153B/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 11.4kB/s]
Device set to use cuda:0
[{'entity': 'I-ORG', 'score': np.float32(0.71511483), 'index': 3, 'word': 'Face', 'start': 8, 'end': 12}, {'entity': 'B-LOC', 'score': np.float32(0.68888517), 'index': 12, 'word': 'NYC', 'start': 12, 'end': 15}, {'entity': 'I-ORG', 'score': np.float32(0.71511483), 'index': 15, 'word': 'and', 'start': 15, 'end': 16}, {'entity': 'I-ORG', 'score': np.float32(0.71511483), 'index': 16, 'word': 'partner', 'start': 16, 'end': 21}, {'entity': 'I-ORG', 'score': np.float32(0.71511483), 'index': 21, 'word': 'with', 'start': 21, 'end': 22}, {'entity': 'I-ORG', 'score': np.float32(0.71511483), 'index': 22, 'word': 'Google', 'start': 22, 'end': 28}, {'entity': 'I-ORG', 'score': np.float32(0.71511483), 'index': 28, 'word': 'with', 'start': 28, 'end': 29}, {'entity': 'I-ORG', 'score': np.float32(0.71511483), 'index': 29, 'word': '$200', 'start': 29, 'end': 33}, {'entity': 'I-ORG', 'score': np.float32(0.71511483), 'index': 33, 'word': 'with', 'start': 33, 'end': 34}, {'entity': 'I-ORG', 'score': np.float32(0.71511483), 'index': 34, 'word': 'dollar', 'start': 34, 'end': 40}, {'entity': 'I-ORG', 'score': np.float32(0.71511483), 'index': 40, 'word': 'price', 'start': 40, 'end': 45}]]
```

```
! pip install transformers datasets
```

```
classifier = pipeline('sentiment-analysis')
result = classifier('I love Hugging Face library very much, it is so nice')
print(result)
```

```
No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f (https://huggingface.co/distilbert/distilbert-base-uncased-finetuned-sst-2-english)
Using a pipeline without specifying a model name and revision in production is not recommended.
```

```
config.json: 100% 629/629 [00:00<00:00, 56.8kB/s]

model.safetensors: 100% 268M/268M [00:06<00:00, 49.6MB/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 3.10kB/s]

vocab.txt: 232k/? [00:00<00:00, 11.3MB/s]
Device set to use cuda:0
[{'label': 'POSITIVE', 'score': 0.999874472618103}]]
```

```
#!pip install gradio
```

```
Requirement already satisfied: gradio in /usr/local/lib/python3.11/dist-packages (5.38.1)
Requirement already satisfied: aiofiles<25.0,>=22.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (24.1.0)
Requirement already satisfied: anyio<5.0,>=3.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (4.9.0)
Requirement already satisfied: brotli<=1.1.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (1.1.0)
Requirement already satisfied: fastapi<1.0,>=0.115.2 in /usr/local/lib/python3.11/dist-packages (from gradio) (0.116.1)
Requirement already satisfied: ffmpeg in /usr/local/lib/python3.11/dist-packages (from gradio) (0.6.1)
Requirement already satisfied: gradio-client==1.11.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (1.11.0)
Requirement already satisfied: groovy~=0.1 in /usr/local/lib/python3.11/dist-packages (from gradio) (0.1.2)
Requirement already satisfied: httpx<1.0,>=0.24.1 in /usr/local/lib/python3.11/dist-packages (from gradio) (0.28.1)
Requirement already satisfied: huggingface-hub<0.28.1 in /usr/local/lib/python3.11/dist-packages (from gradio) (0.33.4)
Requirement already satisfied: jinja2<4.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (3.1.6)
Requirement already satisfied: markupsafe<4.0,>=2.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (3.0.2)
Requirement already satisfied: numpy<3.0,>=1.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (2.0.2)
Requirement already satisfied: orjson<3.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (3.11.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (from gradio) (25.0)
Requirement already satisfied: pandas<3.0,>=1.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (2.2.2)
Requirement already satisfied: pillow<12.0,>=8.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (11.3.0)
```

```

Requirement already satisfied: pydantic<2.12,>=2.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (2.11.7)
Requirement already satisfied: pydub in /usr/local/lib/python3.11/dist-packages (from gradio) (0.25.1)
Requirement already satisfied: python-multipart>=0.0.18 in /usr/local/lib/python3.11/dist-packages (from gradio) (0.0.20)
Requirement already satisfied: pyyaml<7.0,>=5.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (6.0.2)
Requirement already satisfied: ruff>=0.9.3 in /usr/local/lib/python3.11/dist-packages (from gradio) (0.12.4)
Requirement already satisfied: safehttpx<0.2.0,>=0.1.6 in /usr/local/lib/python3.11/dist-packages (from gradio) (0.1.6)
Requirement already satisfied: semantic-version~=2.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (2.10.0)
Requirement already satisfied: starlette<1.0,>=0.40.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (0.47.2)
Requirement already satisfied: tomlkit<0.14.0,>=0.12.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (0.13.3)
Requirement already satisfied: typer<1.0,>=0.12 in /usr/local/lib/python3.11/dist-packages (from gradio) (0.16.0)
Requirement already satisfied: typing-extensions~=4.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (4.14.1)
Requirement already satisfied: uvicorn>=0.14.0 in /usr/local/lib/python3.11/dist-packages (from gradio) (0.35.0)
Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-packages (from gradio-client==1.11.0->gradio) (2025.3.0)
Requirement already satisfied: websockets<16.0,>=10.0 in /usr/local/lib/python3.11/dist-packages (from gradio-client==1.11.0->gradio) (13.1)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.11/dist-packages (from anyio<5.0,>=3.0->gradio) (3.10)
Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.11/dist-packages (from anyio<5.0,>=3.0->gradio) (1.3.1)
Requirement already satisfied: certifi in /usr/local/lib/python3.11/dist-packages (from httpx<1.0,>=0.24.1->gradio) (2025.7.14)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.11/dist-packages (from httpx<1.0,>=0.24.1->gradio) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.11/dist-packages (from httpcore==1.*->httpx<1.0,>=0.24.1->gradio) (0.14.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.28.1->gradio) (3.18.0)
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.28.1->gradio) (2.32.3)
Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.28.1->gradio) (4.67)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.2 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.28.1->gradio) (1.1.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages (from pandas<3.0,>=1.0->gradio) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas<3.0,>=1.0->gradio) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas<3.0,>=1.0->gradio) (2025.2)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.11/dist-packages (from pydantic<2.12,>=2.0->gradio) (0.7.0)
Requirement already satisfied: pydantic-core==2.33.2 in /usr/local/lib/python3.11/dist-packages (from pydantic<2.12,>=2.0->gradio) (2.33.2)
Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.11/dist-packages (from pydantic<2.12,>=2.0->gradio) (0.4.0)
Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.11/dist-packages (from typer<1.0,>=0.12->gradio) (8.2.1)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.11/dist-packages (from typer<1.0,>=0.12->gradio) (1.5.4)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.11/dist-packages (from typer<1.0,>=0.12->gradio) (13.9.4)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.8.2->pandas<3.0,>=1.0->gradio) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.11/dist-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.11/dist-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (2.19.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests->huggingface-hub>=0.28.1->gradio) (3.4.0)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests->huggingface-hub>=0.28.1->gradio) (2.3.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.11/dist-packages (from markdown-it-py>=2.2.0->rich>=10.11.0->typer<1.0,>=0.12->gradio) (0.1.2)

```

```

import gradio as gr

from transformers import pipeline

classifier = pipeline("sentiment-analysis")
def analyze_sentiment(text):
    return classifier(text)[0]['label']

gr.Interface(fn = analyze_sentiment,inputs = 'text',outputs = 'text').launch()

```

 No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f (<https://huggingface.co/distilbert/distilbert-base-uncased-finetuned-sst-2-english>)
Using a pipeline without specifying a model name and revision in production is not recommended.
Device set to use cuda:0
It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting `share=True`

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()

* Running on public URL: <https://09f0c1d3bac9e58f8c.gradio.live>

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the working directory

text I love eating apple	output POSITIVE
Clear	Submit
	Flag